THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An angle is 80° less than its supplement. The angle is: 1

(a) 60° **(b)** 40° **(c)** 70° **(d)** 50°

2. A transversal makes interior angles on the same side of 110° and 70°. The two lines are: 1

(a) equal **(b)** parallel **(c)** perpendicular **(d)** not parallel

3. Two supplementary angles differ by 60°. The larger angle is: 1

(a) 120° **(b)** 130° **(c)** 110° **(d)** 140°

4. Two complementary angles are in the ratio $4 : 5$. The larger angle is: 1

(a) 70° **(b)** 60° **(c)** 50° **(d)** 40°

5. An angle is 17 times its complement. The angle is: 1

(a) 80° **(b)** 85° **(c)** 95° **(d)** 90°

6. Half of the supplement of 120° is: 1

(a) 40° **(b)** 50° **(c)** 20° **(d)** 30°

7. Two interior angles on the same side of a transversal cutting parallel lines are in the ratio $4 : 5$. The smaller angle is: 1

(a) 80° **(b)** 100° **(c)** 70° **(d)** 90°

8. Four angles around a point are 95°, 80°, 130° and x°. Then $x =$ 1

(a) 75° **(b)** 65° **(c)** 45° **(d)** 55°

9. An angle is 120° less than its supplement. The angle is: 1

(a) 30° **(b)** 40° **(c)** 20° **(d)** 50°

10. Two supplementary angles are both acute. This is: 1

(a) possible only for 45° each **(b)** possible only if they are equal

(c) possible **(d)** not possible

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A transversal cuts two lines and makes alternate interior angles of 47° and 47°. Are the lines parallel? Give a reason. 2

12. An angle is 10° more than its complement. Find the angle. 2

13. Two angles forming a linear pair differ by 100°. Find both angles. 2

14. A transversal cuts two lines and makes alternate interior angles of 92° and 88°. Are the lines parallel? Give a reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two lines intersect and two adjacent angles are in the ratio $7 : 5$. Find all four angles. 3

16. An angle is 12° more than twice its complement. Find the angle and its complement. 3

17. Two supplementary angles are in the ratio $5 : 7$. Find both angles and the complement of the smaller one. 3

OR

Two lines intersect and two adjacent angles are in the ratio $1 : 14$. Find all four angles. 3

18. An angle is 24° more than twice its complement. Find the angle and its complement. 3

OR

Two supplementary angles are in the ratio $1 : 19$. Find both angles and the complement of the smaller one. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two interior angles on the same side of a transversal cutting two parallel lines are in the ratio $3 : 7$. Find both angles and the angle forming a linear pair with the larger one. 5

OR

The four angles around a point are in the ratio $2 : 2 : 3 : 5$. Find the angles and say which of them are acute, right or obtuse. 5

20. Two interior angles on the same side of a transversal cutting two parallel lines are in the ratio $1 : 2$. Find both angles and the angle forming a linear pair with the larger one. 5

OR

The four angles around a point are in the ratio $1 : 2 : 3 : 6$. Find the angles and say which of them are acute, right or obtuse. 5

END OF QUESTION PAPER — Total: 40 Marks

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